

Bitcoin Market Sentiment vs Trader Performance Analysis

Introduction

This project analyzes the relationship between Bitcoin market sentiment and trader performance using historical trading data from Hyperliquid along with the Bitcoin Fear & Greed Index.

The objective of this analysis is to understand how market emotions such as Fear, Greed, Extreme Fear, and Extreme Greed influence trading activity, profitability, win rates, and risk-taking behavior.

The project combines two datasets:

1. Bitcoin Fear & Greed sentiment dataset
2. Historical trader transaction dataset from Hyperliquid

The analysis includes:

- Data cleaning and preprocessing
- Dataset merging based on dates
- Profitability analysis
- Win rate analysis
- Trading activity analysis
- Trade size and risk behavior analysis
- Correlation analysis
- Visualization of key insights

The findings from this project help uncover behavioral trading patterns and provide insights into how market sentiment impacts trader decisions and overall market dynamics.

Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')
```

✓ Loading Datasets

Dataset Description

This project uses two datasets:

1. Bitcoin Fear & Greed Index Dataset
 - Contains daily market sentiment classifications such as Fear, Greed, Extreme Fear, and Extreme Greed.
2. Hyperliquid Historical Trader Dataset
 - Contains trader-level transaction data including execution price, trade size, leverage, coin traded, and closed profit/loss values.

These datasets were merged using date information to analyze the relationship between market sentiment and trader performance.

✓ Loading Files

```
trades = pd.read_csv('historical_data.csv')
sentiment = pd.read_csv('fear_greed_index.csv')
```

✓ Check Column Names

```
print(trades.columns)
print(sentiment.columns)
```

```
Index(['Account', 'Coin', 'Execution Price', 'Size Tokens', 'Size USD', 'Side',
      'Timestamp IST', 'Start Position', 'Direction', 'Closed PnL',
      'Transaction Hash', 'Order ID', 'Crossed', 'Fee', 'Trade ID',
      'Timestamp'],
      dtype='object')
Index(['timestamp', 'value', 'classification', 'date'], dtype='object')
```

✓ Preview Data

```
trades.head()
```

	Account	Coin	Execution Price	Size Tokens	Size US
0	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9769	986.87	7872.1
1	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9800	16.00	127.6
2	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9855	144.09	1150.6
3	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9874	142.98	1142.0
4	0xae5eacaf9c6b9111fd53034a602c192a04e082ed	@107	7.9894	8.73	69.7

sentiment.head()

	timestamp	value	classification	date	
0	1517463000	30	Fear	2018-02-01	
1	1517549400	15	Extreme Fear	2018-02-02	
2	1517635800	40	Fear	2018-02-03	
3	1517722200	24	Extreme Fear	2018-02-04	
4	1517808600	11	Extreme Fear	2018-02-05	

Next steps:

[Generate code with sentiment](#)

[New interactive sheet](#)

✓ Check Missing Values

trades.isnull().sum()

	0
Account	0
Coin	0
Execution Price	0
Size Tokens	0
Size USD	0
Side	0
Timestamp IST	0
Start Position	0
Direction	0
Closed PnL	0
Transaction Hash	0
Order ID	0
Crossed	0
Fee	0
Trade ID	0
Timestamp	0

dtype: int64

Check Missing Values

```
sentiment.isnull().sum()
```

	0
timestamp	0
value	0
classification	0
date	0

dtype: int64

✓ Convert Dates

```
trades['Timestamp IST'] = pd.to_datetime(
    trades['Timestamp IST'],
```

```

        format='%d-%m-%Y %H:%M'
    )

    trades['date'] = trades['Timestamp IST'].dt.date

    sentiment['date'] = pd.to_datetime(
        sentiment['date']
    ).dt.date

```

```
trades[['Timestamp IST', 'date']].head()
```

	Timestamp IST	date	
0	2024-12-02 22:50:00	2024-12-02	
1	2024-12-02 22:50:00	2024-12-02	
2	2024-12-02 22:50:00	2024-12-02	
3	2024-12-02 22:50:00	2024-12-02	
4	2024-12-02 22:50:00	2024-12-02	

✓ Merging the Datasets

```

merged = pd.merge(
    trades,
    sentiment[['date', 'classification']],
    on='date',
    how='left'
)

```

✓ Check Merge

```
merged[['date', 'Closed PnL', 'classification']].head()
```

	date	Closed PnL	classification	
0	2024-12-02	0.0	Extreme Greed	
1	2024-12-02	0.0	Extreme Greed	
2	2024-12-02	0.0	Extreme Greed	
3	2024-12-02	0.0	Extreme Greed	
4	2024-12-02	0.0	Extreme Greed	

✓ First Insight

```
merged.groupby('classification')['Closed PnL'].mean()
```

Closed PnL	
classification	
Extreme Fear	34.537862
Extreme Greed	67.892861
Fear	54.290400
Greed	42.743559
Neutral	34.307718

dtype: float64

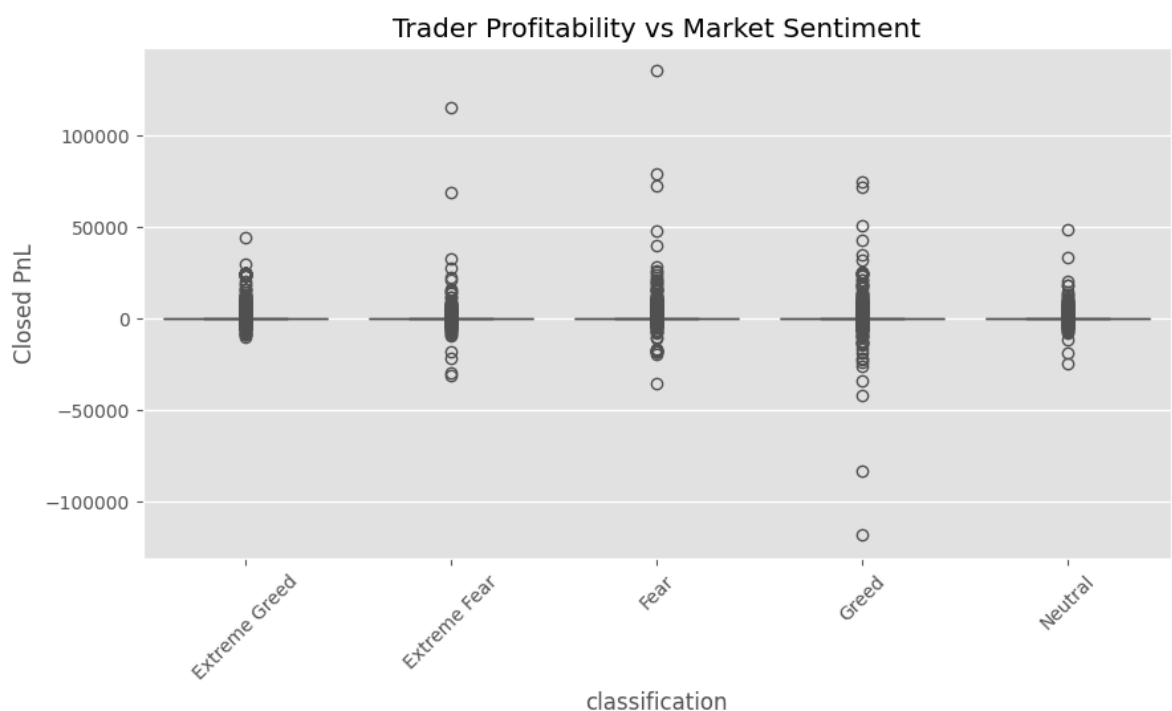
✓ Creating First Visualization

```
plt.figure(figsize=(10,5))

sns.boxplot(
    x='classification',
    y='Closed PnL',
    data=merged
)

plt.title('Trader Profitability vs Market Sentiment')
plt.xticks(rotation=45)

plt.show()
```



Observation

The boxplot shows that trader profitability varies significantly across different market sentiment conditions. Extreme Greed periods produced the highest positive profit outliers, indicating strong bullish momentum and profitable trading opportunities. Fear and Greed markets also showed higher volatility in profit and loss outcomes compared to Neutral conditions.

✓ Trading Activity

```
merged['classification'].value_counts()
```

	count
classification	
Fear	61837
Greed	50303
Extreme Greed	39992
Neutral	37686
Extreme Fear	21400

dtype: int64

✓ Win Rate Analysis

```
merged['profit_trade'] = merged['Closed PnL'] > 0
```

✓ Win Rate Analysis

```
merged.groupby('classification')['profit_trade'].mean() * 100
```

profit_trade	
classification	
Extreme Fear	37.060748
Extreme Greed	46.494299
Fear	42.076750
Greed	38.482794
Neutral	39.699093
dtype: float64	

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Observation

The win-rate analysis indicates that traders had the highest percentage of profitable trades during Extreme Greed periods. In contrast, Extreme Fear conditions showed lower win rates, highlighting the increased uncertainty and risk associated with panic-driven markets.

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Trading Activity

```
activity = merged['classification'].value_counts()

print(activity)
```

```
classification
Fear          61837
Greed         50303
Extreme Greed 39992
Neutral       37686
Extreme Fear  21400
Name: count, dtype: int64
```

Visualization of Trading Activity

```
plt.figure(figsize=(8,5))

sns.countplot(
    x='classification',
```



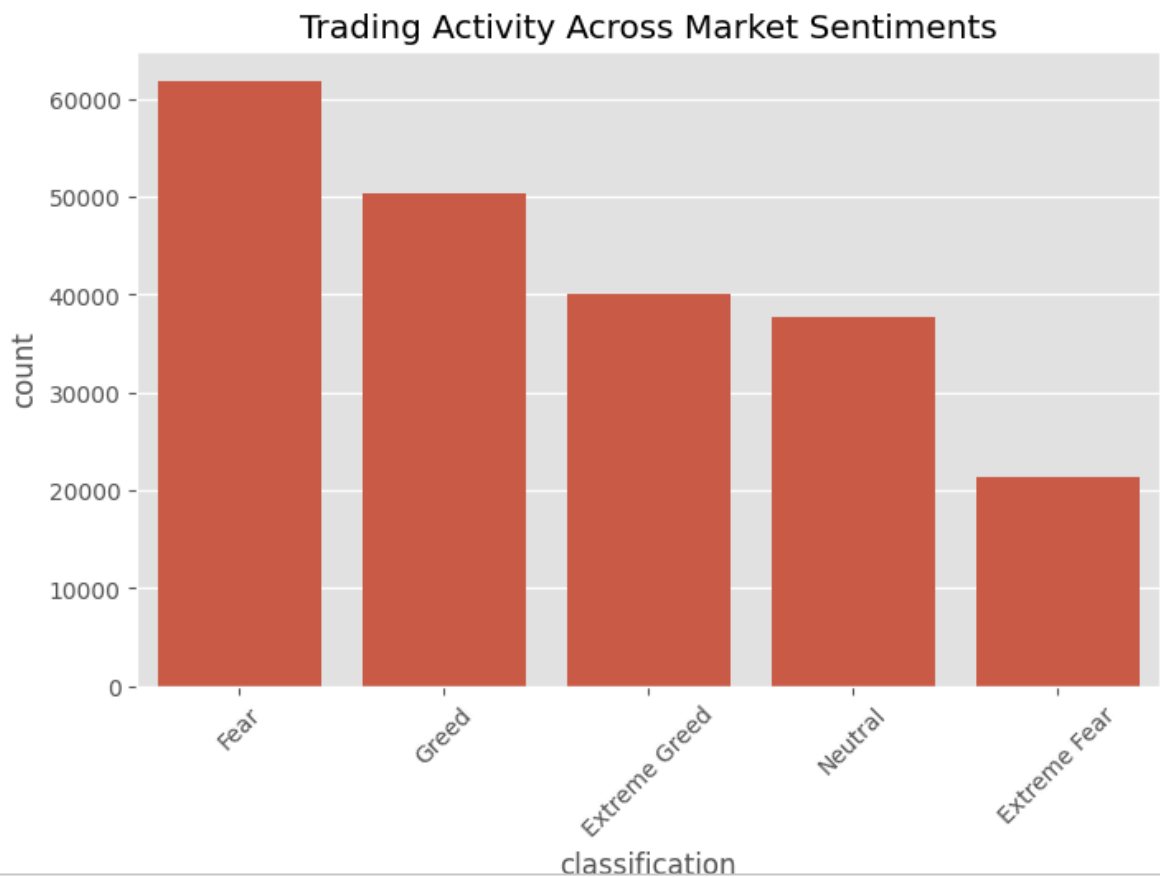
```

data=merged,
order=merged['classification'].value_counts().index
)

plt.title('Trading Activity Across Market Sentiments')
plt.xticks(rotation=45)

plt.show()

```



Observation

The trading activity analysis shows that traders were most active during Fear market conditions. This suggests that volatile and uncertain markets encourage increased participation and short-term trading opportunities. Extreme Fear periods recorded comparatively lower trading activity.

✓ Trade Size and Risk Appetite Analysis

```
merged.groupby('classification')['Size USD'].mean()
```

	Size USD
classification	
Extreme Fear	5349.731843
Extreme Greed	3112.251565
Fear	7816.109931
Greed	5736.884375
Neutral	4782.732661

dtype: float64

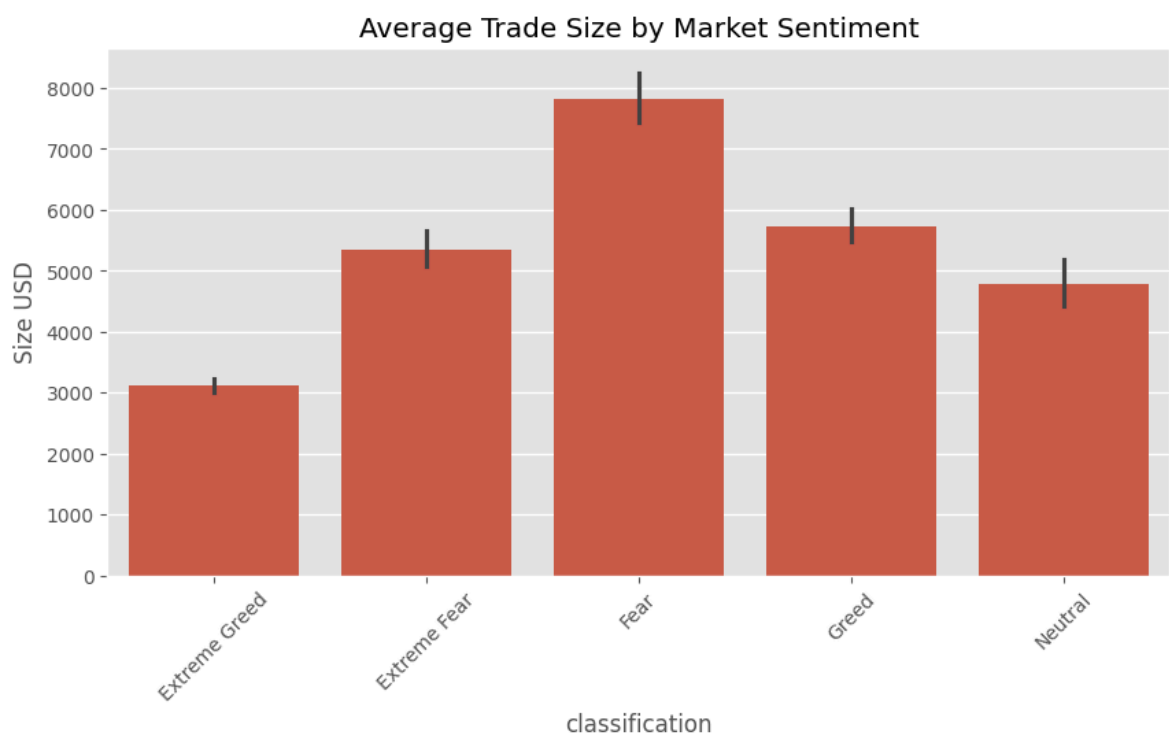
✓ Visualization for Trade Size

```
plt.figure(figsize=(10,5))

sns.barplot(
    x='classification',
    y='Size USD',
    data=merged
)

plt.title('Average Trade Size by Market Sentiment')
plt.xticks(rotation=45)

plt.show()
```



Observation

The analysis reveals that traders placed the largest average trade sizes during Fear market conditions. This indicates increased risk-taking behavior and aggressive positioning during periods of uncertainty and volatility in the crypto market.

✓ Correlation Analysis

```
numeric_cols = ['Execution Price', 'Size USD', 'Closed PnL']

correlation = merged[numeric_cols].corr()

print(correlation)
```

	Execution Price	Size USD	Closed PnL
Execution Price	1.000000	0.189855	-0.006505
Size USD	0.189855	1.000000	0.123589
Closed PnL	-0.006505	0.123589	1.000000

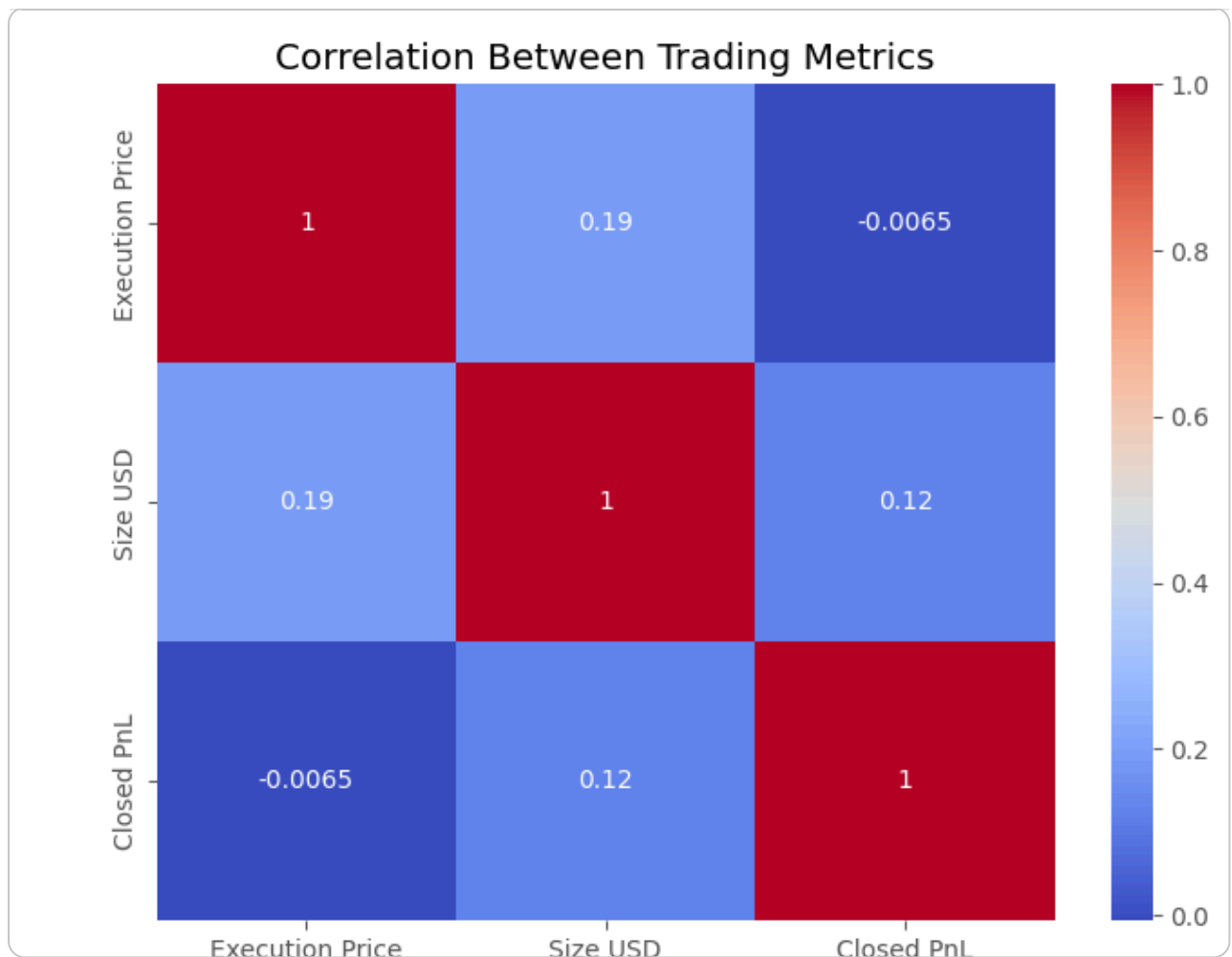
✓ Correlation Heatmap

```
plt.figure(figsize=(8,6))

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm'
)

plt.title('Correlation Between Trading Metrics')

plt.show()
```



Observation

The correlation heatmap helps identify relationships between important trading variables such as execution price, trade size, and closed profit/loss. The analysis suggests that some trading metrics have weak to moderate relationships, highlighting the complex nature of trading performance in crypto markets.

✓ Top Performing Coins

```
top_coins = merged.groupby('Coin')['Closed PnL'].mean().sort_values(ascending=False)
print(top_coins)
```

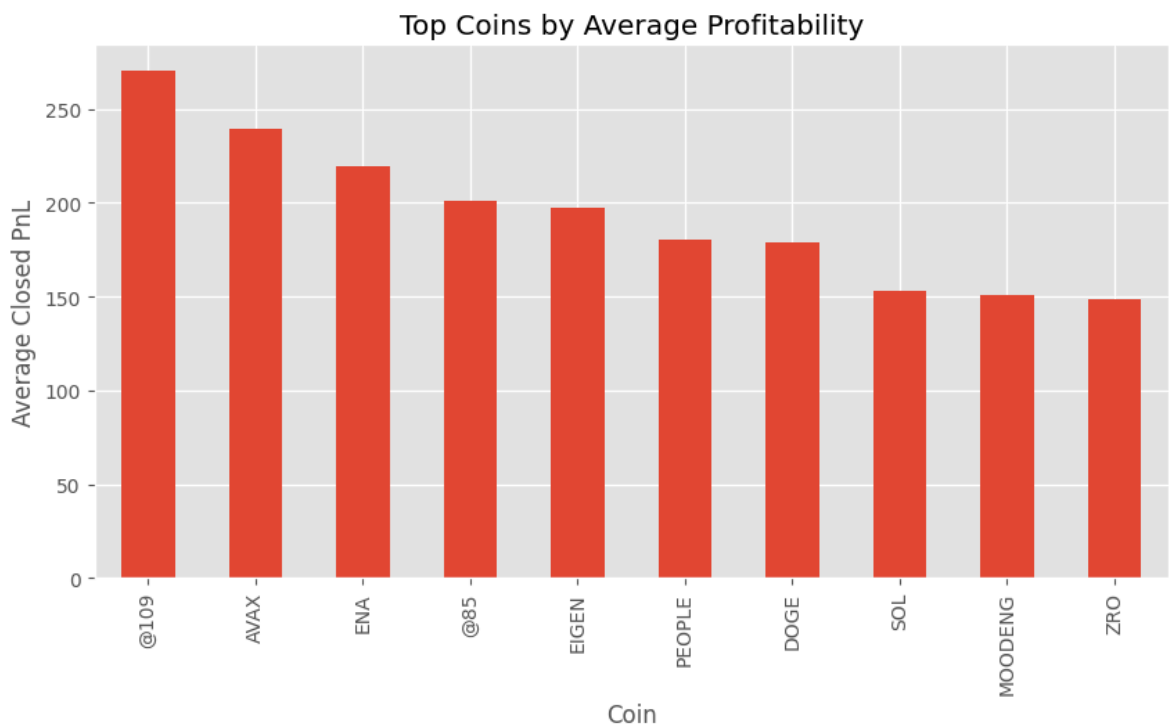
```
Coin
@109      270.704207
AVAX      239.095613
ENA       219.524751
@85       200.804091
EIGEN     197.063189
PEOPLE    180.323003
DOGE      178.623676
SOL       153.358519
MOODENG   151.096714
ZRO       148.327509
Name: Closed PnL, dtype: float64
```

✓ Asset Performance Analysis

```
top_coins.plot(
    kind='bar',
    figsize=(10,5)
)

plt.title('Top Coins by Average Profitability')
plt.ylabel('Average Closed PnL')

plt.show()
```



Observation

The asset performance analysis shows that profitability varied across different cryptocurrencies. Certain coins generated significantly higher average profits, indicating that asset selection plays an important role in overall trading performance.

Key Findings

- Extreme Greed conditions generated the highest trader profitability and win rates.
- Fear markets showed the highest trading activity and largest trade sizes.
- Emotional market conditions created greater volatility in trading outcomes.
- Neutral market conditions produced relatively weaker trading performance.
- Market sentiment significantly influenced trader behavior and risk-taking patterns.

